Topic 7 – Animal coordination, control and homeostasis

Students should:		Maths skills
7.1	Describe where hormones are produced and how they are transported from endocrine glands to their target organs, including the pituitary gland, thyroid gland, pancreas, adrenal glands, ovaries and testes	
7.2	**Explain that adrenalin is produced by the adrenal glands to prepare the body for fight or flight, including:** **a increased heart rate** **b increased blood pressure** **c increased blood flow to the muscles** **d raised blood sugar levels by stimulating the liver to change glycogen into glucose**	2c 4a, 4c
7.3	**Explain how thyroxine controls metabolic rate as an example of negative feedback, including:** **a low levels of thyroxine stimulates production of TRH in hypothalamus** **b this causes release of TSH from the pituitary gland** **c TSH acts on the thyroid to produce thyroxine** **d when thyroxine levels are normal thyroxine inhibits the release of TRH and the production of TSH**	2c 4a, 4c
7.4	Describe the stages of the menstrual cycle, including the roles of the hormones oestrogen and progesterone, in the control of the menstrual cycle	4a
7.5	**Explain the interactions of oestrogen, progesterone, FSH and LH in the control of the menstrual cycle, including the repair and maintenance of the uterus wall, ovulation and menstruation**	4a, 4c
7.6	Explain how hormonal contraception influences the menstrual cycle and prevents pregnancy	
7.7	Evaluate hormonal and barrier methods of contraception	2c, 2d 4a
7.8	**Explain the use of hormones in Assisted Reproductive Technology (ART) including IVF and clomifene therapy**	
7.9	Explain the importance of maintaining a constant internal environment in response to internal and external change	
7.10B	Explain the importance of homeostasis, including: a thermoregulation – the effect on enzyme activity b osmoregulation – the effect on animal cells	

Students should:		Maths skills
7.11B	Explain how thermoregulation takes place, with reference to the function of the skin, including: a the role of the dermis b the role of the epidermis c the role of the hypothalamus	
7.12B	Explain how thermoregulation takes place, with reference to: a shivering **b vasoconstriction** **c vasodilation**	
7.13	Explain how the hormone insulin controls blood glucose concentration	
7.14	**Explain how blood glucose concentration is regulated by glucagon**	
7.15	Explain the cause of type 1 diabetes and how it is controlled	
7.16	Explain the cause of type 2 diabetes and how it is controlled	
7.17	Evaluate the correlation between body mass and type 2 diabetes including waist:hip calculations and BMI, using the BMI equation: $\text{BMI} = \frac{\text{mass (kg)}}{(\text{height (m)})^2}$	1a, 1c, 2c 2e, 3a
7.18B	Describe the structure of the urinary system	
7.19B	Explain how the structure of the nephron is related to its function in filtering the blood and forming urine including: a filtration in the glomerulus and Bowman's capsule b selective reabsorption of glucose c reabsorption of water	
7.20B	**Explain the effect of ADH on the permeability of the collecting duct in regulating the water content of the blood**	
7.21B	Describe the treatments for kidney failure, including kidney dialysis and organ donation	
7.22B	State that urea is produced from the breakdown of excess amino acids in the liver	

Use of mathematics

- Use simple compound measures such as rate (1a, 1c).
- Plot, draw and interpret appropriate graphs (4a, 4b, 4c and 4d).
- Translate information between numerical and graphical forms (4a).
- Construct and interpret frequency tables and diagrams, bar charts and histograms (2c).
- Understand and use percentiles (1c).
- Extract and interpret data from graphs, charts and tables (1c).

Suggested practical

- Investigate the presence of sugar in simulated urine/body fluids.